Blind Separation of BPSK, QPSK, 8PSK, $\pi/4$-DQPSK and MSK by Zero-Lag Cyclic Autocorrelation

Abstract

We show that the zero-lag, all-unconjugated cyclic autocorrelations of orders 2, 4 and 8 blindly separate the five classes BPSK, QPSK, 8PSK, $\pi/4$-DQPSK and MSK, without knowledge of carrier offset, carrier phase, timing offset, symbol rate, amplitude, or pulse shape (within stated conditions), in additive circular noise. Plain differentially encoded QPSK is shown to be statistically identical to QPSK and is therefore excluded. The underlying cycle-frequency structure and cyclic-feature framework are established in [1, 2, 3]; the blind "type" classification (Definition 2), the uniqueness-of-center argument (Lemma 4.9) and the peak-symmetry test (Theorem 2) are, to our knowledge, new.

1 Signal model

The observation is $y(t) = s(t)+w(t)$, $t \in \mathbb{R}$, where s belongs to $\mathcal{C} = \{\text{BPSK, QPSK, 8PSK, } \pi/4\text{-DQPSK, MSK}\}$ or $s \equiv 0$. For the linear classes (M-PSK and $\pi/4$-DQPSK) [4],

$$s(t) = A\, e^{j(2\pi f_0 t+\varphi)} \sum_{k\in\mathbb{Z}} a_k\, p(t - kT - t_0). \tag{1}$$

MSK is written in its OQPSK form [5, 1],

$$s(t) = A\, e^{j(2\pi f_0 t+\varphi)} \Big[\sum_k a_k\, q(t - 2kT_b - t_0) + j \sum_k b_k\, q\big(t - (2k+1)T_b - t_0\big) \Big]. \tag{2}$$

We write $\widehat{f}(\nu) = \int_{\mathbb{R}} f(t)e^{-j2\pi\nu t}\, dt$, $G_n = \widehat{p^n}$ and $Q_2 = \widehat{q^2}$.

2 Assumptions

Assumption A1 (Static parameters)**.** $A > 0$, $f_0 \in \mathbb{R}$, $\varphi \in [0, 2\pi)$, $t_0 \in \mathbb{R}$, $T > 0$, $T_b > 0$ are unknown deterministic constants (static flat channel, with its complex gain absorbed into A and φ; stable carrier and symbol clock).

Assumption A2 (Symbols)**.**

(a) *M-PSK, $M \in \{2, 4, 8\}$:* a_k i.i.d. uniform on $\{e^{j(\theta_c+2\pi m/M)} : m = 0, \dots, M-1\}$, with θ_c a fixed constant.

(b) *$\pi/4$-DQPSK* [6]: $a_k = e^{j\psi_k}$ with $\psi_k = \sum_{i=1}^{k} \Delta_i$ for $k \geq 0$ and $\psi_k = -\sum_{i=k+1}^{0} \Delta_i$ for $k < 0$, where Δ_i are i.i.d. uniform on $\mathcal{D} = \{\pm\pi/4, \pm 3\pi/4\}$. (The initial phase is absorbed into φ.)

(c) *MSK:* a_k, b_k i.i.d. uniform on $\{\pm 1\}$, mutually independent.

Assumption A3 (Pulses)**.** p and q are real-valued, measurable, bounded, not a.e. zero, and

$$S_p := \sup_t \sum_k |p(t - kT)| < \infty, \qquad S_q := \sup_t \sum_k |q(t - kT_b)| < \infty.$$

Consequently $p, q \in L^1 \cap L^\infty$ (e.g. $\int |p| \le T S_p$), and $p^n, q^2 \in L^1 \cap L^\infty$ for all $n \ge 1$.

Assumption A4 (Non-degeneracy)**.** For $\pi/4$-DQPSK: $G_4\big(\frac{2m+1}{2T}\big) \neq 0$ for some $m \in \mathbb{Z}$. For MSK: $Q_2\big(\frac{2m+1}{2T_b}\big) \neq 0$ for some $m \in \mathbb{Z}$.

Assumption A5 (Noise)**.** w is independent of s; all unconjugated moments vanish, $\mathbb{E}[w(t_1) \cdots w(t_m)] = 0$ for all $m \ge 1$ and all t_i; and $\sup_t \mathbb{E}\,|w(t)|^8 < \infty$. (Circular complex Gaussian noise satisfies this [7, 8].)

Assumption A6 (Closed set)**.** Exactly one class of $\mathcal{C}$ is present, or $s \equiv 0$; there are no co-channel signals.

3 Definitions

Definition 1 (Zero-lag CAF)**.** For $n \in \mathbb{N}$ let $m_n(t) = \mathbb{E}[y^n(t)]$ and

$$R_n^\alpha = \lim_{L\to\infty} \frac{1}{2L} \int_{-L}^{L} m_n(t)\, e^{-j2\pi\alpha t}\, dt, \qquad M_n(\alpha) = |R_n^\alpha|\,.$$

(Existence is established in Lemma 4.7. For $n = 2$ this is the zero-lag conjugate cyclic autocorrelation [9, 1]; for $n > 2$ it is a zero-lag cyclic moment [2]. Zero-lag higher-order features are used for classification in [10, 11].)

Definition 2 (Type)**.** The type $\tau_n \in \{\varnothing, \mathrm{C}, \mathrm{O}\}$ is defined as follows. If $M_n \equiv 0$, then $\tau_n = \varnothing$. Otherwise let c_n be the unique point about which M_n is symmetric (Lemmas 4.8 and 4.9); then $\tau_n = \mathrm{C}$ ("centered") if $M_n(c_n) > 0$ and $\tau_n = \mathrm{O}$ ("offset") if $M_n(c_n) = 0$.

4 Lemmas

Lemma 4.1 (Periodic sums)**.** *Let $f \in L^1$ with $\sup_t \sum_k |f(t - kP)| < \infty$, and set $F(t) = \sum_k f(t - kP - t_0)$ and $F_{\mathrm{alt}}(t) = \sum_k (-1)^k f(t - kP - t_0)$. Then*

$$\lim_{L\to\infty} \frac{1}{2L} \int_{-L}^{L} F(t) e^{-j2\pi\alpha t}\, dt = \begin{cases} \frac{1}{P}\widehat{f}(\alpha) e^{-j2\pi\alpha t_0}, & \alpha \in \frac{1}{P}\mathbb{Z}, \\ 0, & \textit{otherwise}, \end{cases}$$

and the same limit for F_{alt} equals $\frac{1}{P}\widehat{f}(\alpha)e^{-j2\pi\alpha t_0}$ if $\alpha \in \frac{1}{P}(\mathbb{Z} + \frac{1}{2})$ and 0 otherwise. If merely $f \in L^1$, then $\frac{1}{2L}\int_{-L}^{L} f(t) e^{-j2\pi\alpha t} dt \to 0$.

Proof. The last claim follows since the integral is bounded by $\|f\|_1$. F and F_{alt} are bounded; F is P-periodic and F_{alt} is $2P$-periodic with $F_{\mathrm{alt}}(t + P) = -F_{\mathrm{alt}}(t)$. For a bounded P'-periodic G, split $[-L, L]$ into whole periods plus a remainder contributing $O(1/L)$. The whole periods contribute $\frac{1}{P'}\int_0^{P'} G(t) e^{-j2\pi\alpha t} dt$ times an average of $e^{-j2\pi\alpha n P'}$ over n, which equals 1 if $\alpha \in \mathbb{Z}/P'$ and is otherwise a normalized bounded geometric sum tending to 0. The limit is therefore the Fourier coefficient on $\mathbb{Z}/P'$ and 0 elsewhere [12, 13]. For F and $\alpha \in \mathbb{Z}/P$, unfold the sum (Fubini, $f \in L^1$) using $e^{-j2\pi\alpha k P} = 1$: $\frac{1}{P}\int_0^P F(t) e^{-j2\pi\alpha t} dt = \frac{1}{P}\widehat{f}(\alpha) e^{-j2\pi\alpha t_0}$. For F_{alt} and $\alpha \in \mathbb{Z}/(2P)$, write $k = 2l + r$, $r \in \{0, 1\}$, and unfold each sub-lattice to get $\frac{1}{2P}\widehat{f}(\alpha) e^{-j2\pi\alpha t_0}\big(1 - e^{-j2\pi\alpha P}\big)$. With $\alpha = i/(2P)$ the bracket is $1 - (-1)^i$, which is 0 for even i and 2 for odd i. □

Lemma 4.2 (M-PSK moments)**.** *Under A2(a),* $\mathbb{E}[a^r] = e^{jr\theta_c} \mathbb{1}[M \mid r]$.

Proof. $\mathbb{E}[a^r] = e^{jr\theta_c} \frac{1}{M} \sum_{m=0}^{M-1} e^{j2\pi mr/M}$, and the geometric sum equals $\mathbb{1}[M \mid r]$ [14]. □

Lemma 4.3 (Products of i.i.d. symbols)**.** *Let* (a_k) *be i.i.d. and* $n \geq 1$.

(a) If $\mathbb{E}[a^r] = 0$ *for all* $1 \leq r \leq n$*, then* $\mathbb{E}[a_{k_1} \cdots a_{k_n}] = 0$ *for every index tuple.*

(b) If $\mathbb{E}[a^r] = 0$ *for* $1 \leq r < n$*, then* $\mathbb{E}[a_{k_1} \cdots a_{k_n}] = \mathbb{E}[a^n] \mathbb{1}[k_1 = \cdots = k_n]$.

Proof. Group equal indices into blocks of sizes r_b with $\sum_b r_b = n$. By independence the expectation is $\prod_b \mathbb{E}[a^{r_b}]$. In (a) every $r_b \leq n$, so every factor vanishes. In (b) the product vanishes unless there is a single block, $r = n$. (Cf. the partition structure of moments and cumulants [2, 3].) □

Lemma 4.4 (Products of $\pi/4$-DQPSK symbols)**.** *Under A2(b), for* $n \in \{2, 4\}$*,*

$$\mathbb{E}[a_{k_1} \cdots a_{k_n}] = \mathbb{1}[k_1 = \cdots = k_n = k] \cdot \begin{cases} \mathbb{1}[k = 0], & n = 2, \\ (-1)^k, & n = 4. \end{cases}$$

Proof. By the phase recursion, $\sum_\ell \psi_{k_\ell} = \sum_i c_i \Delta_i$ with $c_i = \#\{\ell : k_\ell \geq i\}$ for $i \geq 1$ and $c_i = -\#\{\ell : k_\ell < i\}$ for $i \leq 0$; only finitely many c_i are nonzero. By independence the expectation factors as $\prod_i \mathbb{E}[e^{jc_i\Delta}]$. Since $\mathcal{D}$ is symmetric,

$$\mathbb{E}[e^{jc\Delta}] = \tfrac{1}{2}\Big[\cos\tfrac{c\pi}{4} + \cos\tfrac{3c\pi}{4}\Big] = \begin{cases} 0, & |c| \in \{1, 2, 3\}, \\ -1, & |c| = 4. \end{cases}$$

If the indices are not all equal, take $i = \max_\ell k_\ell$. Since $\min_\ell k_\ell < i$, we have $1 \leq |c_i| \leq n - 1 \leq 3$, so the factor for Δ_i is 0 and the expectation vanishes. If all indices equal k, exactly $|k|$ coefficients are nonzero, each of magnitude n, giving $0^{|k|} = \mathbb{1}[k = 0]$ for $n = 2$ and $(-1)^{|k|} = (-1)^k$ for $n = 4$. □

Lemma 4.5 (Moments of s)**.** *Let* $\theta = \varphi + \theta_c$ *for M-PSK and* $\theta = \varphi$ *otherwise.*

$$\begin{aligned}
\textit{BPSK:}\quad & \mathbb{E}[s^2] = A^2 e^{j2\theta} e^{j4\pi f_0 t} \textstyle\sum_k p^2(t - kT - t_0). \\
\textit{QPSK:}\quad & \mathbb{E}[s^2] = 0, \qquad \mathbb{E}[s^4] = A^4 e^{j4\theta} e^{j8\pi f_0 t} \textstyle\sum_k p^4(t - kT - t_0). \\
\textit{8PSK:}\quad & \mathbb{E}[s^2] = \mathbb{E}[s^4] = 0, \qquad \mathbb{E}[s^8] = A^8 e^{j8\theta} e^{j16\pi f_0 t} \textstyle\sum_k p^8(t - kT - t_0). \\
\pi/4\textit{-DQPSK:}\quad & \mathbb{E}[s^2] = A^2 e^{j2\theta} e^{j4\pi f_0 t}\, p^2(t - t_0), \\
& \mathbb{E}[s^4] = A^4 e^{j4\theta} e^{j8\pi f_0 t} \textstyle\sum_k (-1)^k p^4(t - kT - t_0). \\
\textit{MSK:}\quad & \mathbb{E}[s^2] = A^2 e^{j2\theta} e^{j4\pi f_0 t} \textstyle\sum_k (-1)^k q^2(t - kT_b - t_0).
\end{aligned}$$

(Unit-modulus constants $\mathbb{E}[a^M] e^{-jM\theta_c}$ *are absorbed into* $e^{jM\theta}$*.)*

Proof. The n-fold sum for s^n is dominated by $(AS_p)^n$ since $|a_k| = 1$, so expectation and summation can be exchanged by dominated convergence. For M-PSK apply Lemma 4.3 with Lemma 4.2: for BPSK, $\mathbb{E}[a] = 0$ and $\mathbb{E}[a^2] = e^{j2\theta_c}$; for QPSK, $\mathbb{E}[a^r] = 0$ for $r \leq 3$ and $\big|\mathbb{E}[a^4]\big| = 1$; for 8PSK, $\mathbb{E}[a^r] = 0$ for $r \leq 7$ and $\big|\mathbb{E}[a^8]\big| = 1$ [3, 11]. For $\pi/4$-DQPSK apply Lemma 4.4. For MSK write $s = Ae^{j(2\pi f_0 t + \varphi)}(I + jQ)$, so that $(I + jQ)^2 = I^2 - Q^2 + 2jIQ$. Since $\mathbb{E}[a_k a_\ell] = \delta_{k\ell}$, $\mathbb{E}[I^2] = \sum_k q^2(t - 2kT_b - t_0)$, and similarly $\mathbb{E}[Q^2]$ on the odd shifts; $\mathbb{E}[IQ] = \mathbb{E}[I]\,\mathbb{E}[Q] = 0$. Combining the even and odd shifts gives the alternating sum with spacing T_b (cf. [1] for the half-sine case). The exchanges are justified by S_q. □

Lemma 4.6 (Noise invariance)**.** *Under A5, $m_n(t) = \mathbb{E}[s^n(t)]$ for $n \le 8$.*

Proof. By the binomial theorem and independence, $\mathbb{E}[(s+w)^n] = \sum_m \binom{n}{m} \mathbb{E}[s^{n-m}]\,\mathbb{E}[w^m]$. Every term with $m \ge 1$ vanishes; the moments are finite because s is bounded and $\mathbb{E}\,|w|^8$ is bounded [7]. □

Lemma 4.7 (CAF magnitudes)**.** *The limits R_n^α exist for $n \in \{2, 4, 8\}$. With $\Lambda_P = \frac{1}{P}\mathbb{Z}$ and $\Lambda'_P = \frac{1}{P}(\mathbb{Z} + \frac{1}{2})$, and writing $\alpha = nf_0 + \nu$:*

Class	$M_n(nf_0 + \nu)$
BPSK	$M_2 = A^2 \,\lvert G_2(\nu)\rvert / T$ *on* Λ_T
QPSK	$M_2 \equiv 0$; $M_4 = A^4 \,\lvert G_4(\nu)\rvert / T$ *on* Λ_T
8PSK	$M_2 \equiv M_4 \equiv 0$; $M_8 = A^8 \,\lvert G_8(\nu)\rvert / T$ *on* Λ_T
$\pi/4$*-DQPSK*	$M_2 \equiv 0$; $M_4 = A^4 \,\lvert G_4(\nu)\rvert / T$ *on* Λ'_T
MSK	$M_2 = A^2 \,\lvert Q_2(\nu)\rvert / T_b$ *on* Λ'_{T_b}
$s \equiv 0$	$M_n \equiv 0$ *for all* n

Each M_n is zero off the stated lattice.

Proof. Combine Lemmas 4.5 and 4.6 and apply Lemma 4.1 with $f = p^n$ or $f = q^2$; its hypotheses hold because $|p|^n \le \|p\|_\infty^{n-1}\,|p|$ (similarly for q). The factor $e^{j2\pi n f_0 t}$ shifts α by nf_0. The second-order $\pi/4$-DQPSK term is an L^1 transient and so contributes 0 in the limit. The integer lattice for PSK agrees with [3, 11]. □

Lemma 4.8 (Symmetry and square-summability)**.** *Each nonzero M_n in Lemma 4.7 is symmetric about nf_0, and $\sum_\alpha M_n(\alpha)^2 < \infty$.*

Proof. For real $f \in L^1$, $\widehat{f}(-\nu) = \overline{\widehat{f}(\nu)}$, and each lattice Λ_P, Λ'_P is symmetric about 0. The values of M_n are moduli of Fourier coefficients of a bounded periodic function, hence square-summable by Parseval. □

Lemma 4.9 (Unique center)**.** *Let $M \ge 0$ be nonzero, with countable support and $\sum M^2 < \infty$. Then M is symmetric about at most one point.*

Proof. If M were symmetric about $c_1 \ne c_2$, composing the two reflections would make M invariant under translation by $d = 2(c_2 - c_1) \ne 0$. Choosing α_0 with $M(\alpha_0) > 0$ gives $M(\alpha_0 + kd) = M(\alpha_0)$ for all $k \in \mathbb{Z}$, so $\sum M^2 = \infty$, a contradiction. □

Lemma 4.10 (Center values)**.** *For BPSK ($n = 2$), QPSK ($n = 4$) and 8PSK ($n = 8$), $M_n(nf_0) = A^n\big(\int p^n\big)/T > 0$. For $\pi/4$-DQPSK ($n = 4$) and MSK ($n = 2$), $M_n(nf_0) = 0$, and under A4 $M_n \not\equiv 0$.*

Proof. In the first group $0 \in \Lambda_T$ and $G_n(0) = \int p^n > 0$, because n is even and p is real and not a.e. zero. In the second group $0 \notin \Lambda'_P$, so the center value is 0, and A4 supplies a nonzero value elsewhere on the lattice. □

5 Main results

Theorem 1 (Population-level separability)**.** *Under A1–A6, the types* (τ_2, τ_4, τ_8) *are:*

Class	τ_2	τ_4	τ_8
BPSK	C	–	–
MSK	O	–	–
QPSK	$\varnothing$	C	–
$\pi/4$*-DQPSK*	$\varnothing$	O	–
8PSK	$\varnothing$	$\varnothing$	C
$s \equiv 0$	$\varnothing$	$\varnothing$	$\varnothing$

(A dash means the value is not needed.) The class-to-type map is injective on $\mathcal{C} \cup \{s \equiv 0\}$*, and* (τ_2, τ_4) *alone is injective on* $\mathcal{C}$*. The types are invariant to* A*,* f_0*,* φ*,* t_0*,* T*,* T_b*, the noise distribution, and the pulses within A3 and A4. Hence the zero-lag CAFs of orders* 2*,* 4 *and* 8 *blindly separate the five classes.*

Proof. By Lemma 4.7, each M_n is either identically zero or nonzero. A nonzero M_n is symmetric about nf_0 and square-summable (Lemma 4.8), so by Lemma 4.9 $c_n = nf_0$ is its unique center and the type is well defined. Lemma 4.10 decides between C and O. Reading off Lemmas 4.7 and 4.10: BPSK has $M_2 \not\equiv 0$ with $M_2(2f_0) > 0$; MSK has $M_2 \not\equiv 0$ with $M_2(2f_0) = 0$; QPSK has $M_2 \equiv 0$ and $M_4(4f_0) > 0$; $\pi/4$-DQPSK has $M_2 \equiv 0$, $M_4 \not\equiv 0$ and $M_4(4f_0) = 0$; 8PSK has $M_2 \equiv M_4 \equiv 0$ and $M_8(8f_0) > 0$; and every M_n vanishes for $s \equiv 0$ by Lemma 4.6. The rows are pairwise distinct, and none of the defining quantities depends on the parameters listed. □

Theorem 2 (Peak-symmetry test)**.** *Under A1–A6, for* $n \in \{2, 4, 8\}$ *with* $M_n \not\equiv 0$*, the maximum of* M_n *is attained, and* $\tau_n = \mathrm{C}$ *if and only if* M_n *is symmetric about a maximizer; in that case the maximizer is unique and equals* nf_0*.*

Proof. Square-summability implies that only finitely many values exceed any $\varepsilon > 0$, so the supremum is attained. *Case C.* Since n is even and $p^n \geq 0$ is not a.e. zero, for $\nu \neq 0$ we have $|G_n(\nu)| \leq \int p^n$, with equality only if $e^{-j2\pi\nu t}$ is a.e. constant on $\mathrm{supp}(p^n)$, which is impossible on a set of positive measure. Hence $|G_n(\nu)| < G_n(0)$, the unique maximizer is nf_0, and M_n is symmetric about it (Lemma 4.8). *Case O.* Any maximizer α^* satisfies $M_n(\alpha^*) > 0 = M_n(nf_0)$, so $\alpha^* \neq nf_0$. Since M_n is symmetric about nf_0, Lemma 4.9 rules out symmetry about α^*. □

Assumption A7 (Mixing)**.** For $n \in \{2, 4, 8\}$, the process $z(t) = y^n(t)$ has cumulants up to order 2 (with any conjugation pattern) that are absolutely integrable in their lag differences, uniformly in t [2, 15].

Theorem 3 (Consistency)**.** *Under A1–A7, let* $\hat{R}_n^\alpha(L) = \frac{1}{L}\int_0^L y^n(t)e^{-j2\pi\alpha t}\,dt$*. Then* $\mathbb{E}\left|\hat{R}_n^\alpha(L) - R_n^\alpha\right|^2 = O(1/L)$ *for each fixed* α*. Line/no-line decisions on a finite candidate set, with thresholds* $\eta_L \to 0$ *and* $\sqrt{L}\,\eta_L \to \infty$*, are consistent, and the classification error probability tends to* 0*.*

Proof sketch. The bias is $O(1/L)$ by the period-splitting argument of Lemma 4.1. The variance is L^{-2} times a double integral of the covariance of z, which is $O(L)$ by A7. Chebyshev's inequality gives consistency of each thresholded decision, and a union bound extends this to a finite candidate set. Uniform extensions over a continuous search range in α follow [15]; see also [9, 3]. □

6 Exclusion of plain DQPSK

Proposition 1. *If $\psi_k = \psi_{k-1} + \Delta_k$ with Δ_k i.i.d. uniform on $\{0, \pi/2, \pi, 3\pi/2\}$, then (a_k) has the same joint law as i.i.d. uniform QPSK. Hence plain DQPSK and QPSK are the same random process, and no statistic of y separates them.*

Proof. Δ_k is independent of $(\psi_0, \ldots, \psi_{k-1})$ and uniform on the group $\frac{\pi}{2}\mathbb{Z} \bmod 2\pi$. Adding an independent uniform group element yields a uniform element independent of the past; induction completes the proof (cf. differential encoding in [4]). □

7 Remarks

Remark 1 (Channel). A dispersive or time-varying channel makes the effective pulse complex or time-varying, violating A1 and A3. Lemmas 4.8 and 4.10 may then fail, but the lattice supports of Lemma 4.7 survive. Separability can be recovered by estimating f_0 and T and testing lattice membership [11, 16], by using nonzero lags or the full spectral correlation function [1, 17, 18], or by equalizing first.

Remark 2 (Discrete time). The results are stated in continuous time. After sampling, the rate must be high enough that the lines of y^4 and y^8 do not alias onto one another; this holds for bandlimited pulses, and only approximately (after prefiltering) for rectangular pulses.

Remark 3 (Half-sine MSK). For $q(t) = \cos\big(\pi t/(2T_b)\big)$ on $|t| < T_b$, one has $\left|Q_2(\pm\frac{1}{2T_b})\right| = T_b/2$ and $Q_2 = 0$ at all other odd half-multiples of $1/T_b$, so $M_2 = A^2/2$ at $2f_0 \pm \frac{1}{2T_b}$ and zero elsewhere, consistent with [1].

Remark 4 (Provenance). The cycle-frequency lattices, the cyclic-moment and cumulant framework, and power-law features are established in [1, 2, 3, 11, 10, 14]; the offset fourth-order lines of $\pi/4$-DQPSK are discussed in [19]. Definition 2, Lemma 4.9, Theorem 2, the increment-coefficient proof of Lemma 4.4, and the general-pulse MSK form in Lemma 4.5 are, to our knowledge, new.

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